



Overparameterized Models

UMC 205 : Introduction to AI and ML

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Models today are **massively overparameterized**

More parameters than training data.





- Modern machine learning models often operate in a regime where the number of parameters d exceeds the number of training samples n
- Statistical learning theory predicts that such models should overfit and exhibit poor generalization
- Deep neural networks achieve near-zero training error

Why do overparameterized models generalize well, even though they can perfectly fit the training data?



This was the central puzzle of modern deep learning.

- Training data:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n, \quad (x_i, y_i) \sim \mathcal{P}$$

- Model parameterized by $w \in \mathbb{R}^d$, prediction: $f_w(x)$
- Population (true) risk:

$$L(w) = \mathbb{E}_{(x,y) \sim \mathcal{P}} [\ell(f_w(x), y)]$$

- Optimal predictor:

$$w^* = \arg \min_w L(w)$$

- Empirical risk:

$$\hat{L}(w) = \frac{1}{n} \sum_{i=1}^n \ell(f_w(x_i), y_i)$$

- Generalization gap:

$$L(w) - \hat{L}(w)$$

How do overparameterized models learn the training set perfectly?

What is the nature of this training process?



Trying to understand what's happening

Understanding this is key to explain why they generalize well on unseen data.

- CIFAR-10 experiments with label randomization of level p .
- $1-p$ fraction of labels unchanged, and p fraction of labels is replaced with a random choice over 10 classes.
- Randomization for both train and test data

Question What would you expect the OPM to behave like (say at $p=0.5$)?

- Bayes predictor: Risk= 0.45
- 1-NN classifier (Simple Interpolator): Risk=0.675
- Observed result on Overparameterized model- Risk=0.51

OPM $\neq$ Blind Interpolation



Figure: Test error vs epochs for $p=0.5$

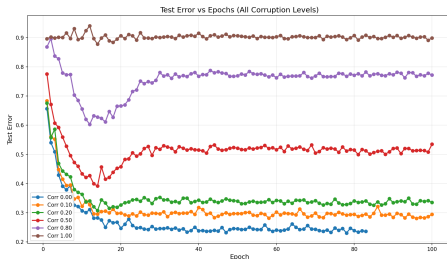


Figure: Test error vs epochs for different values of p

- At all levels of randomization we can verify that the OPM gives near optimal performance
- This indicates that it is learning the signal rather than just blindly interpolating

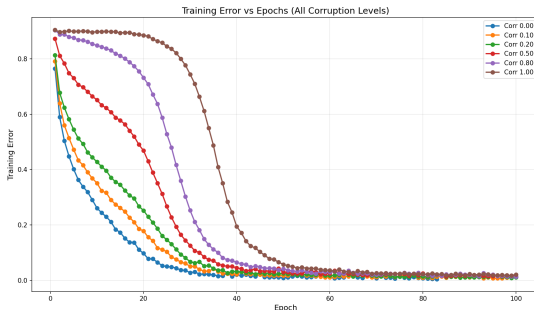


Figure: Train error vs epochs for different values of p

- Training process has remained largely unchanged
- Over-fitting and learning noise doesn't adversely affect the training process

Thus we conclude :

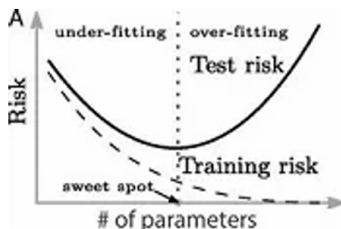
- OPMs predominantly learn the signal present in data
- They learn microscopic noise to be able to fit to train data perfectly.
- It is this global learning of the signal that allows the models to generalize well.

We now view how this idea manifests in the form of lower risk as we proceed in the direction of increasing overparameterization. We discuss this behavior in the context of bias variance trade-off

- **Generalization vs Model Complexity** – classical learning theory predicts a tradeoff

Here, model complexity is defined in terms of number of parameters

Optimal performance is achieved at an intermediate model complexity



- Classical view:

Test error $\uparrow$ beyond a certain number of parameters

- In modern models:

Test error decreases again in highly overparameterized regime

This contradicts the bias–variance tradeoff

Double Descent replaces classical view of bias-variance tradeoff

- **Modern Observation** – test error does not always follow U-shape

Instead, it exhibits a “double descent” curve as model complexity increases

- **Interpolation Threshold**

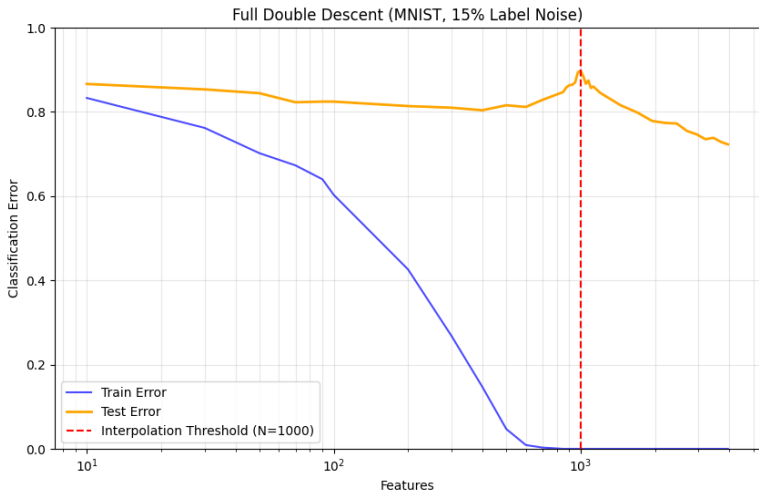
- Point where model can perfectly fit training data
- Classical theory predicts worst generalization here

- **Second Descent**

- Increasing parameters beyond interpolation improves generalization
- Overparameterized models can perform better

- **Implication** – more parameters $\nrightarrow$ worse performance

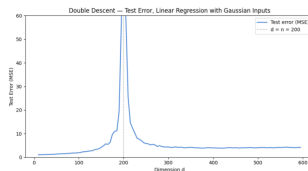
Result: Double descent Behavior



Experiment – Varying Input Dimension (d)



Setup: $n = 200$ samples, Gaussian noise ($\sigma = 1.0$). We varied dimension d from 10 to 600.



Results:

- For $d \leq n - 2$:

$$\mathbb{E}R(\hat{\theta}) = \sigma^2 \frac{d}{n - d - 1}$$

- For $d \geq n + 2$:

$$\mathbb{E}R(\hat{\theta}) = \sigma^2 \frac{n}{d - n - 1} + \|\theta^*\|^2 \frac{d - n}{d}$$

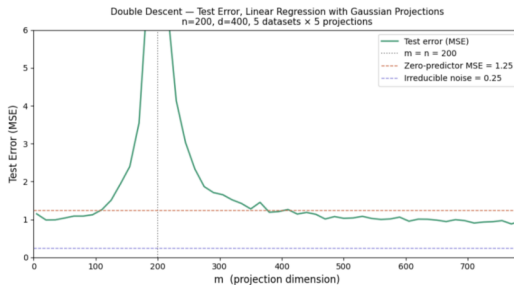


Figure: Gaussian projection of data in high dimensions



• Key Observations

- Training loss $\rightarrow 0$ as model size increases
- Test error initially decreases, then increases, peaking at the interpolating threshold
- Beyond interpolating threshold, test error **decreases again**
- The second descent in the overparameterized regime is referred to as the Double descent

• Conclusion

- Overparameterized models can generalize well
- Classical U-shaped curve is incomplete to explain the behavior beyond the interpolation regime.



- Overparameterized models:
 - Can interpolate training data
 - Many solutions achieve zero training error
- However:
 - Not all solutions generalize equally well

Problem

What determines which solution is selected among many that fit the data?

- **Common Approach:**

- Regularization is used to bias models toward simpler or more stable solutions
- Helps select among many interpolating solutions

- **Widely Used Techniques:**

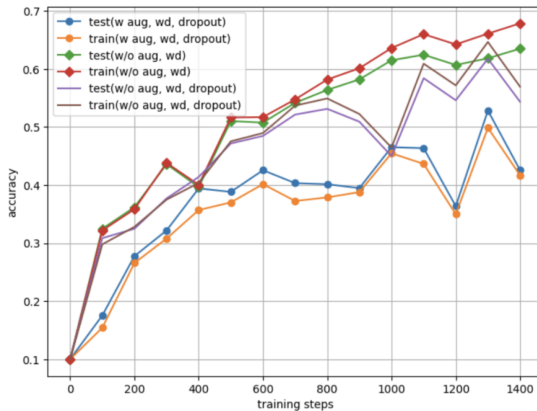
- Data augmentation
- Weight decay (ℓ_2 regularization)
- Dropout

- **Intuition:**

- Designed to control overfitting
- Encourage better generalization to unseen data

Key Question

Are these explicit regularization techniques the primary reason for good generalization in overparameterized models?



ResNet18 on CIFAR-10 trained with/without data augmentation, weight decay, and dropout.

- **Training behavior:**
 - All models achieve high training accuracy even without regularization
- **Test performance:**
 - Explicit regularization provides only modest improvement
- **Key Insight:**
 - Explicit regularization is not the primary reason for generalization

Conclusion

Generalization in overparameterized models is not driven by explicit regularization as we thought.

- If explicit regularization is not the key to this problem then what could it be?

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Think inside the box!

- It is believed that that the implicit regularization could be the key to it.
- Theoretically it is shown that **Stochastic Gradient Descent** has some implicit biases in the overparameterized regimes.



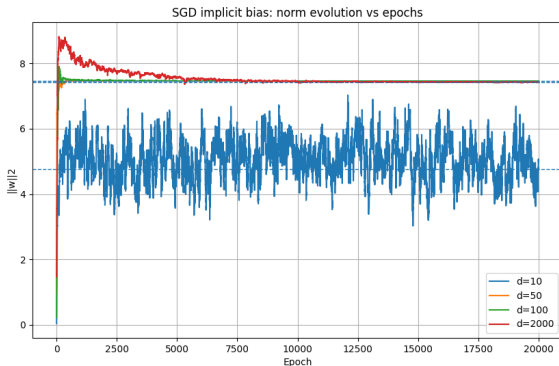


Figure: SGD implicit bias: evolution of $\|w\|_2$ over epochs

$$y = X\theta \text{ where } X \in \mathbb{R}^{1000 \times 50} \sim \mathcal{N}(0, I) \text{ and } \theta \in \mathbb{R}^{50} \sim \mathcal{N}(0, I)$$



- In the overparameterized regime ($d > n$), the system $y = Xw$ has infinitely many solutions.
- **Implicit Bias:** SGD converges to the solution with the smallest Euclidean norm.

Closed-Form Solution

For the noiseless case $y = X\theta$, the minimum norm solution w^* is:

$$w^* = X^\top (XX^\top)^{-1} y$$

- Since $y = X\theta^*$, we have $w^* = X^\top (XX^\top)^{-1} X\theta$.
- This w^* is the orthogonal projection of θ onto the row space of X .

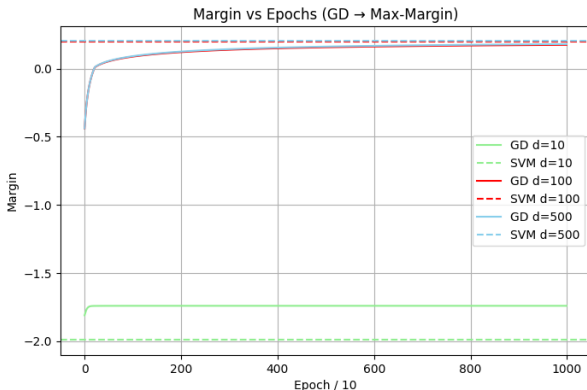


Figure: SGD implicit bias: evolution of Margin over epochs

$$y = \text{sign}(Xw) \text{ where } X \in \mathbb{R}^{100 \times 20} \sim \mathcal{N}(0, I) \text{ and } w \in \mathbb{R}^{20} \sim \mathcal{N}(0, I)$$

Takeaway: Max-Margin Bias



- In the overparameterized regime, infinitely many classifiers can perfectly separate the data ($y_i(w^\top x_i) > 0$).

Observations for Logistic Loss

- **Norm Growth:** $\|w_t\| \rightarrow \infty$ as the optimizer pushes toward zero loss.
- **Directional Convergence:** The weight vector direction $\frac{w_t}{\|w_t\|}$ stabilizes.

Implicit Bias of SGD: The direction converges to the **Hard-Margin SVM** solution:

$$\lim_{t \rightarrow \infty} \frac{w_t}{\|w_t\|} = \arg \max_{\|w\|=1} \min_i y_i(w^\top x_i)$$

SGD implicitly maximizes the margin without any explicit ℓ_2 penalty.



Comparing different optimizers

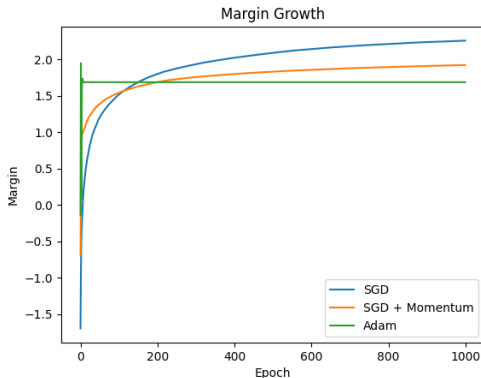


Figure: Growth of margin according to different optimizers

Conclusion:

- Adam Found a solution almost immediately but plateaus at a suboptimal margin (1.7), failing to show the long-term growth behavior of SGD.
- Momentum provides early speed, but vanilla SGD shows a stronger implicit bias toward the **maximum-margin solution** over time.

- In overparameterized models, **implicit bias of optimization** selects specific solutions

But what determines this bias?

Not just optimization...

Also the choice of architecture

(especially activation functions)

Effect of Activation functions on Generalization : Motivation



- Generalization in deep neural networks is not fully explained by capacity alone.
- Activation functions strongly influence:
 - Optimization dynamics
 - Gradient propagation
 - Final performance

Key Question

How do different activation functions affect generalization?

Margin Definition

$$\gamma(x, y) = f_y(x) - \max_{j \neq y} f_j(x)$$

- Train Accuracy / Validation Accuracy
- Test Accuracy
- Generalization Gap
- Margin Distribution



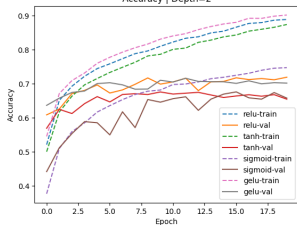
Just
Accuracy

Accuracy
+ Margin +
Generalization

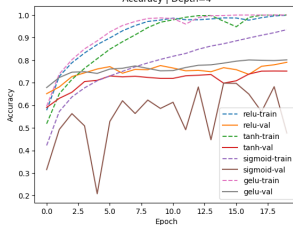
Results: Accuracy Behaviour



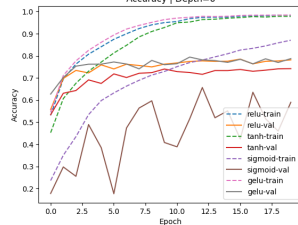
Accuracy | Depth=2



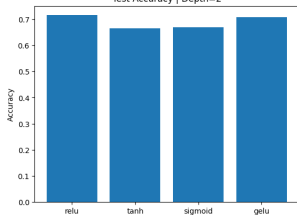
Accuracy | Depth=4



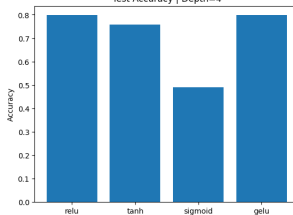
Accuracy | Depth=6



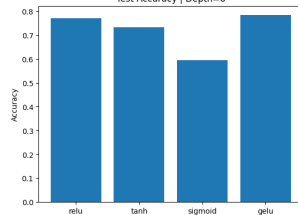
Test Accuracy | Depth=2



Test Accuracy | Depth=4



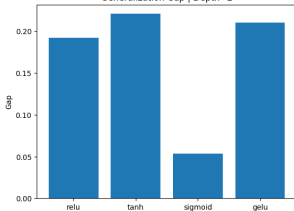
Test Accuracy | Depth=6



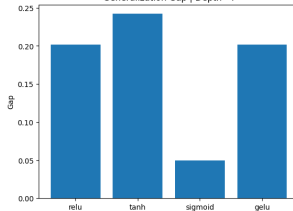
Results: Generalization Gap Across Depths



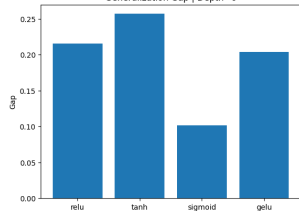
Generalization Gap | Depth=2



Generalization Gap | Depth=4



Generalization Gap | Depth=6



Key Insight

Small gap $\neq$ good generalization (Sigmoid underfits)

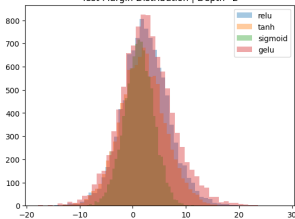
Conclusion

Generalization gap alone is not a reliable metric.

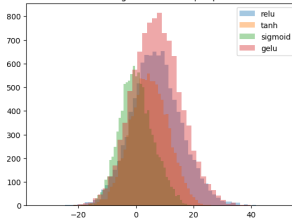
Results: Margin Distributions Across Depths



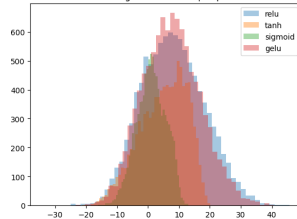
Test Margin Distribution | Depth=2



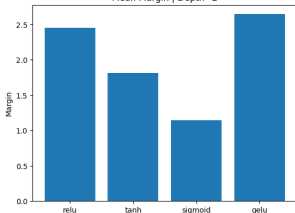
Test Margin Distribution | Depth=4



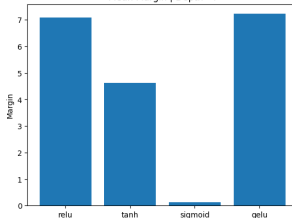
Test Margin Distribution | Depth=6



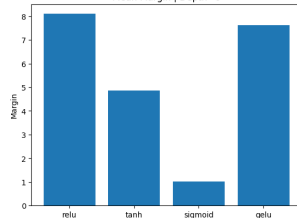
Mean Margin | Depth=2



Mean Margin | Depth=4



Mean Margin | Depth=6



Why Does This Happen?



- Core reason: Gradient propagation
- Sigmoid:

$$\sigma'(x) \leq 0.25 \Rightarrow (0.25)^L \rightarrow 0$$

- Vanishing gradients $\rightarrow$ no learning
- Tanh:
 - Partial gradient decay $\rightarrow$ slower learning
- ReLU / GELU:
 - Gradients preserved $\rightarrow$ efficient optimization

Pipeline

Activation $\rightarrow$ Optimization $\rightarrow$ Margins $\rightarrow$ Generalization

- Activation functions impact generalization **indirectly**
- Key driver: **Optimization dynamics**
- Observations:
 - ReLU, GELU: Best performance and margins
 - Tanh: Moderate
 - Sigmoid: Fails due to vanishing gradients
 - Generalization gap alone is not sufficient
 - Margins correlate better with performance

- Overparameterized models can perfectly fit even random labels
 - $\Rightarrow$ Capacity alone does not explain generalization
- Double descent shows:
 - Test error decreases beyond interpolation
- Many interpolating solutions exist
 - $\Rightarrow$ Generalization depends on solution selection
- Solution selection is governed by implicit bias
- Activation functions influence this process via optimization
 - $\Rightarrow$ Affect margins and generalization

Final Insight

Generalization in deep learning is determined not just by model size, but by how optimization and architecture shape the learned solution.



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